

ROS1 fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes ROS1 gene fusions from the msk_impact_50k_2026 study, covering 122 fusion events. 72/118 fusions (61.0%) were in-frame. 107/118 fusions (90.7%) retained the Protein kinase domain. Domain retention was tested with Fisher's exact test ($p=0.214047$, not statistically significant at $\alpha=0.05$). No literature benchmark is configured for ROS1.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 122 analyzed fusion events, identifying 30 distinct fusion partner genes. The most frequent partner was CD74, observed in 55 events.

Domain retention

Protein kinase domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 67/72 (93.1%) of in-frame fusions retained the domain, $p=0.214047$ (not statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.002997.

Domain disruption

Disruption of the Fibronectin type III domain was tested with Fisher's exact test comparing in-frame fusions against all others; 68/72 (94.4%) of in-frame fusions disrupted the domain, $p=0.0770779$ (not statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.966103.

Cutpoint detection

Cutpoint detection scanned 118 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 1693 aa (exon 30; permutation-corrected $p=0.0959041$, not statistically significant at $\alpha=0.05$). This is 25 aa from the nearest configured domain boundary at 1668 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" ($n=107$) and "not_retained" ($n=11$). For Frame_status, retained: 67/90 (74.4%, 95% CI 64.6%-82.3%); not_retained: 5/9 (55.6%, 95% CI 26.7%-81.1%). Welch's t-test comparing Tumor_variant_count between groups gave means of 68.5514 vs 19.0909, $p=3.23984e-07$ (statistically significant at $\alpha=0.05$).

Composite evidence score

Composite evidence score produced results for this run (30 row(s) in composite_evidence_ranking); see the accompanying table.

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Expression association

Expression association reported: No mRNA expression data was available for ROS1 in this cohort; expression-association analysis was skipped.

Joint partner

Joint partner reported: ROS1 has no gene_pair configured; joint-partner analysis was skipped.

Mutation cooccurrence

Mutation cooccurrence reported: ROS1 has no mutual_exclusivity_targets configured; co-occurrence/mutual-exclusivity analysis was skipped.

Window detection

Window detection produced results for this run (71 row(s) in window_scan, 40 row(s) in top_windows); see the accompanying table.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
CD74	55	0.45081967213114754	0.25233128227596563	0.0014976394126361699	0.9185841308626892	0.6389004592256574	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.400305915642832	1
SDC4	17	0.13934426229508196	0.25233128227596563	0.0014976394126361699	0.9540766825972906	0.6389004592256574	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.31218717545220254	2
EZR	13	0.10655737704918032	0.25233128227596563	0.0014976394126361699	0.9188947887787228	0.6389004592256574	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2970738258056469	3
COL13A1	1	0.00819672131147541	0.25233128227596563	0.0014976394126361699	1.0	0.6389004592256574	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.279731410767527	4
ZNF157	1	0.00819672131147541	0.25233128227596563	0.0014976394126361699	1.0	0.6389004592256574	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.279731410767527	5
AFG1L	1	0.00819672131147541	0.25233128227596563	0.0014976394126361699	0.9993891264508247	0.6389004592256574	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2796397797351507	6
SLC34A2	1	0.00819672131147541	0.25233128227596563	0.0014976394126361699	0.9993891264508247	0.6389004592256574	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2796397797351507	7
FN1	1	0.00819672131147541	0.25233128227596563	0.0014976394126361699	0.9651802076970067	0.6389004592256574	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.274508441922078	8
RBPJL	1	0.00819672131147541	0.25233128227596563	0.0014976394126361699	0.9651802076970067	0.6389004592256574	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.274508441922078	9

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
NKAIN2	1	0.00819672131147541	0.25233128227596563	0.0014976394126361699	0.9639584605986561	0.6389004592256574	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2743251798573254	10
STX7	1	0.00819672131147541	0.25233128227596563	0.0014976394126361699	0.9413561392791692	0.6389004592256574	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.27093483165940235	11
GOPC	4	0.03278688524590164	0.25233128227596563	0.0014976394126361699	0.8782834453268173	0.6389004592256574	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.26885097674687747	12
SLC4A4	2	0.01639344262295082	0.25233128227596563	0.0014976394126361699	0.9022602321319487	0.6389004592256574	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.26752946198076194	13
TFG	3	0.02459016393442623	0.25233128227596563	0.0014976394126361699	0.8851557727550396	0.6389004592256574	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2674228094676682	14
IGF1R	1	0.00819672131147541	0.25233128227596563	0.0014976394126361699	0.9022602321319487	0.6389004592256574	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2650704455873193	15
LARS1	1	0.00819672131147541	0.25233128227596563	0.0014976394126361699	0.9022602321319487	0.6389004592256574	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2650704455873193	16
LRIG3	2	0.01639344262295082	0.25233128227596563	0.0014976394126361699	0.8851557727550396	0.6389004592256574	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2649637930742256	17
NUDCD3	2	0.01639344262295082	0.25233128227596563	0.0014976394126361699	0.8851557727550396	0.6389004592256574	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2649637930742256	18
TPM3	2	0.01639344262295082	0.25233128227596563	0.0014976394126361699	0.8851557727550396	0.6389004592256574	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2649637930742256	19

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
CCDC30	1	0.00819672131147541	0.25233128227596563	0.0014976394126361699	0.8851557727550396	0.6389004592256574	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2625047766807829	20
GOLGB1	1	0.00819672131147541	0.25233128227596563	0.0014976394126361699	0.8851557727550396	0.6389004592256574	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2625047766807829	21
MYH9	1	0.00819672131147541	0.25233128227596563	0.0014976394126361699	0.8576664630421502	0.6389004592256574	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.25838138022384954	22
PRKN	1	0.00819672131147541	0.25233128227596563	0.0014976394126361699	0.8393402565668906	0.6389004592256574	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.25563244925256057	23
CEP85L	2	0.01639344262295082	0.25233128227596563	0.0014976394126361699	0.7486255345143555	0.6389004592256574	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.24448425733812296	24
GRIK2	1	0.00819672131147541	0.25233128227596563	0.0014976394126361699	0.6731826511912034	0.6389004592256574	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.2307088084462075	25
NUS1	1	0.00819672131147541	0.25233128227596563	0.0014976394126361699	0.2229688454489921	0.6389004592256574	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.16317673758487583	26
CERT1	1	0.00819672131147541	0.25233128227596563	0.0014976394126361699	None	0.6389004592256574	confidence_certainty, domain_disruption, domain_retention, recurrence	0.15262518913826706	27
EYS	1	0.00819672131147541	0.25233128227596563	0.0014976394126361699	None	0.6389004592256574	confidence_certainty, domain_disruption, domain_retention, recurrence	0.15262518913826706	28
NETO1	1	0.00819672131147541	0.25233128227596563	0.0014976394126361699	None	0.6389004592256574	confidence_certainty, domain_disruption, domain_retention, recurrence	0.15262518913826706	29
SLC16A10	1	0.00819672131147541	0.25233128227596563	0.0014976394126361699	0.0	0.6389004592256574	confidence_certainty, cutpoint_proximity, domain_disruption, domain_retention, recurrence	0.129731410767527	30

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
56	0	107	1	10	0.0	0.09322033898305085	1.0304893221479003
421	1	106	1	10	0.09433962264150944	0.17847312762567	0.7484271655598992
875	2	105	1	10	0.19047619047619047	0.25637653793702886	0.5911217213946672
1430	3	104	1	10	0.28846153846153844	0.32750573865609567	0.4847810857642878
1634	4	103	1	10	0.3883495145631068	0.39239553580331443	0.4062759421356547
1688	5	102	1	10	0.49019607843137253	0.4515428730260006	0.34530100806444014
1693	5	102	3	8	0.13071895424836602	0.026435659744030223	1.577809846612513
1694	8	99	3	8	0.21548821548821548	0.06612721119379524	1.1796197925642002
1750	26	81	4	7	0.5617283950617284	0.4675112851720896	0.3302079013073733
1755	27	80	4	7	0.590625	0.47622392260409285	0.3221887920600884
1789	44	63	6	5	0.582010582010582	0.5242960133693759	0.28042344455265833
1800	45	62	6	5	0.6048387096774194	0.527867316527827	0.27747522697617977
1840	47	60	6	5	0.6527777777777778	0.5392135184056008	0.2682392283136832
1853	91	16	10	1	0.56875	1.0	-0.0
1881	104	3	11	0	0.0	1.0	-0.0
1926	106	1	11	0	0.0	1.0	-0.0

domain_disruption tables

frame_domain_contingency_table

68	39
4	7

domain_disruption_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
fibronectin	Fibronectin type III domain	118	11	0	107	0.0

domain_retention tables

frame_domain_contingency_table

67	40
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5	6
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domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein kinase domain	118	107	0	11	0.0

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
36	107	122	0.8770491803278688

frequency tables

Partner_gene_counts

Partner_gene	Event_count
AFG1L	1
CCDC30	1
CD74	55
CEP85L	2
CERT1	1
COL13A1	1
EYS	1
EZR	13
FN1	1
GOLGB1	1
GOPC	4
GRIK2	1
IGF1R	1
LARS1	1
LRIG3	2
MYH9	1
NETO1	1
NKAIN2	1
NUDCD3	2

Partner_gene	Event_count
NUS1	1
PRKN	1
RBPJL	1
SDC4	17
SLC16A10	1
SLC34A2	1
SLC4A4	2
STX7	1
TFG	3
TPM3	2
ZNF157	1

window_detection tables

window_scan

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
1493	1693	200	4	114	2	105	0.08571428571428572	0.042988935779828465	1.366643306003459	EVT-P-0007870-T02-IM6-207, EVT-P-0037864-T02-IM7-49, EVT-P-0054667-T01-IM6-30, EVT-P-0055135-T03-IM7-90
1494	1694	200	7	111	5	102	0.22058823529411764	0.12834492320203683	0.8916213055585486	EVT-P-0007870-T02-IM6-207, EVT-P-0016046-T01-IM6-171, EVT-P-0037022-T01-IM6-1, EVT-P-0037864-T02-IM7-49, EVT-P-0054497-T01-IM6-8, ... (7 total)
1550	1750	200	26	92	23	84	0.7301587301587301	0.7049161533582773	0.15186253733397728	EVT-P-0001840-T01-IM3-185, EVT-P-0001840-T02-IM5-186, EVT-P-0004414-T01-IM5-190, EVT-P-0004946-T02-IM5-192, EVT-P-0007870-T02-IM6-207, ... (26 total)
1555	1755	200	27	91	24	83	0.7710843373493976	0.7124856955112606	0.14722385050947034	EVT-P-0001117-T01-IM3-100, EVT-P-0001840-T01-IM3-185, EVT-P-0001840-T02-IM5-186, EVT-P-0004414-T01-IM5-190, EVT-P-0004946-T02-IM5-192, ... (27 total)
1589	1789	200	46	72	41	66	0.7454545454545455	0.7487650178578418	0.1256544539416864	EVT-P-0000569-T01-IM3-76, EVT-P-0001117-T01-IM3-100, EVT-P-0001437-T01-IM3-74, EVT-P-0001840-T01-IM3-185, EVT-P-0001840-T02-IM5-186, ... (46 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
1593	1693	100	4	114	2	105	0.08571428571428572	0.042988935779828465	1.366643306003459	EVT-P-0007870-T02-IM6-207, EVT-P-0037864-T02-IM7-49, EVT-P-0054667-T01-IM6-30, EVT-P-0055135-T03-IM7-90
1594	1694	100	7	111	5	102	0.22058823529411764	0.12834492320203683	0.8916213055585486	EVT-P-0007870-T02-IM6-207, EVT-P-0016046-T01-IM6-171, EVT-P-0037022-T01-IM6-1, EVT-P-0037864-T02-IM7-49, EVT-P-0054497-T01-IM6-8, ... (7 total)
1600	1800	200	47	71	42	65	0.7753846153846153	0.7520920961097245	0.12372897538802877	EVT-P-0000569-T01-IM3-76, EVT-P-0001117-T01-IM3-100, EVT-P-0001437-T01-IM3-74, EVT-P-0001840-T01-IM3-185, EVT-P-0001840-T02-IM5-186, ... (47 total)
1634	1734	100	7	111	5	102	0.22058823529411764	0.12834492320203683	0.8916213055585486	EVT-P-0007870-T02-IM6-207, EVT-P-0016046-T01-IM6-171, EVT-P-0037022-T01-IM6-1, EVT-P-0037864-T02-IM7-49, EVT-P-0054497-T01-IM6-8, ... (7 total)
1634	1834	200	47	71	42	65	0.7753846153846153	0.7520920961097245	0.12372897538802877	EVT-P-0000569-T01-IM3-76, EVT-P-0001117-T01-IM3-100, EVT-P-0001437-T01-IM3-74, EVT-P-0001840-T01-IM3-185, EVT-P-0001840-T02-IM5-186, ... (47 total)
1640	1840	200	48	70	43	64	0.80625	0.7562186041130197	0.12135264254576777	EVT-P-0000569-T01-IM3-76, EVT-P-0001117-T01-IM3-100, EVT-P-0001437-T01-IM3-74, EVT-P-0001840-T01-IM3-185, EVT-P-0001840-T02-IM5-186, ... (48 total)
1644	1694	50	6	112	4	103	0.17475728155339806	0.09665884968988354	1.0147583775843583	EVT-P-0007870-T02-IM6-207, EVT-P-0016046-T01-IM6-171, EVT-P-0037022-T01-IM6-1, EVT-P-0054497-T01-IM6-8, EVT-P-0054667-T01-IM6-30, ... (6 total)
1650	1750	100	25	93	22	85	0.6901960784313725	0.698430794091426	0.15587662066395394	EVT-P-0001840-T01-IM3-185, EVT-P-0001840-T02-IM5-186, EVT-P-0004414-T01-IM5-190, EVT-P-0004946-T02-IM5-192, EVT-P-0007870-T02-IM6-207, ... (25 total)
1653	1853	200	96	22	87	20	0.9666666666666667	1.0	-0.0	EVT-P-0000569-T01-IM3-76, EVT-P-0000632-T01-IM3-70, EVT-P-0000840-T01-IM3-78, EVT-P-0000840-T03-IM6-22, EVT-P-0001117-T01-IM3-100, ... (96 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
1655	1755	100	26	92	23	84	0.7301587301587301	0.7049161533582773	0.1518625373339728	EVT-P-0001117-T01-IM3-100, EVT-P-0001840-T01-IM3-185, EVT-P-0001840-T02-IM5-186, EVT-P-0004414-T01-IM5-190, EVT-P-0004946-T02-IM5-192, ... (26 total)
1669	1694	25	6	112	4	103	0.17475728155339806	0.09665884968988354	1.0147583775843583	EVT-P-0007870-T02-IM6-207, EVT-P-0016046-T01-IM6-171, EVT-P-0037022-T01-IM6-1, EVT-P-0054497-T01-IM6-8, EVT-P-0054667-T01-IM6-30, ... (6 total)
1681	1881	200	110	8	100	7	1.4285714285714286	0.5544227008978931	0.25615899579731366	EVT-P-0000569-T01-IM3-76, EVT-P-0000632-T01-IM3-70, EVT-P-0000840-T01-IM3-78, EVT-P-0000840-T03-IM6-22, EVT-P-0001117-T01-IM3-100, ... (110 total)
1688	1713	25	6	112	4	103	0.17475728155339806	0.09665884968988354	1.0147583775843583	EVT-P-0007870-T02-IM6-207, EVT-P-0016046-T01-IM6-171, EVT-P-0037022-T01-IM6-1, EVT-P-0054497-T01-IM6-8, EVT-P-0054667-T01-IM6-30, ... (6 total)
1688	1738	50	6	112	4	103	0.17475728155339806	0.09665884968988354	1.0147583775843583	EVT-P-0007870-T02-IM6-207, EVT-P-0016046-T01-IM6-171, EVT-P-0037022-T01-IM6-1, EVT-P-0054497-T01-IM6-8, EVT-P-0054667-T01-IM6-30, ... (6 total)
1688	1788	100	26	92	23	84	0.7301587301587301	0.7049161533582773	0.1518625373339728	EVT-P-0001117-T01-IM3-100, EVT-P-0001840-T01-IM3-185, EVT-P-0001840-T02-IM5-186, EVT-P-0004414-T01-IM5-190, EVT-P-0004946-T02-IM5-192, ... (26 total)
1688	1888	200	110	8	100	7	1.4285714285714286	0.5544227008978931	0.25615899579731366	EVT-P-0000569-T01-IM3-76, EVT-P-0000632-T01-IM3-70, EVT-P-0000840-T01-IM3-78, EVT-P-0000840-T03-IM6-22, EVT-P-0001117-T01-IM3-100, ... (110 total)
1689	1789	100	44	74	39	68	0.6882352941176471	0.7446835754012113	0.12802822476719072	EVT-P-0000569-T01-IM3-76, EVT-P-0001117-T01-IM3-100, EVT-P-0001437-T01-IM3-74, EVT-P-0001840-T01-IM3-185, EVT-P-0001840-T02-IM5-186, ... (44 total)
1693	1718	25	5	113	3	104	0.12980769230769232	0.06794655006722032	1.1678325893333363	EVT-P-0007870-T02-IM6-207, EVT-P-0016046-T01-IM6-171, EVT-P-0037022-T01-IM6-1, EVT-P-0054497-T01-IM6-8, EVT-P-0055135-T03-IM7-90

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
1693	1743	50	5	113	3	104	0.12980769230769232	0.06794655006722032	1.1678325893333363	EVT-P-0007870-T02-IM6-207, EVT-P-0016046-T01-IM6-171, EVT-P-0037022-T01-IM6-1, EVT-P-0054497-T01-IM6-8, EVT-P-0055135-T03-IM7-90
1693	1793	100	44	74	39	68	0.6882352941176471	0.7446835754012113	0.12802822476719072	EVT-P-0000569-T01-IM3-76, EVT-P-0001117-T01-IM3-100, EVT-P-0001437-T01-IM3-74, EVT-P-0001840-T01-IM3-185, EVT-P-0001840-T02-IM5-186, ... (44 total)
1693	1893	200	109	9	99	8	1.2375	1.0	-0.0	EVT-P-0000569-T01-IM3-76, EVT-P-0000632-T01-IM3-70, EVT-P-0000840-T01-IM3-78, EVT-P-0000840-T03-IM6-22, EVT-P-0001117-T01-IM3-100, ... (109 total)
1694	1794	100	42	76	39	68	1.5294117647058822	0.7442597575635433	0.12827546285762348	EVT-P-0000569-T01-IM3-76, EVT-P-0001117-T01-IM3-100, EVT-P-0001437-T01-IM3-74, EVT-P-0001840-T01-IM3-185, EVT-P-0001840-T02-IM5-186, ... (42 total)
1694	1894	200	107	11	99	8	4.640625	0.06612721119379524	1.1796197925642002	EVT-P-0000569-T01-IM3-76, EVT-P-0000632-T01-IM3-70, EVT-P-0000840-T01-IM3-78, EVT-P-0000840-T03-IM6-22, EVT-P-0001117-T01-IM3-100, ... (107 total)
1700	1750	50	19	99	18	89	2.0224719101123596	1.0	-0.0	EVT-P-0001840-T01-IM3-185, EVT-P-0001840-T02-IM5-186, EVT-P-0004414-T01-IM5-190, EVT-P-0004946-T02-IM5-192, EVT-P-0012008-T01-IM5-198, ... (19 total)
1700	1800	100	40	78	37	70	1.4095238095238096	0.7476634359140739	0.12629385777653632	EVT-P-0000569-T01-IM3-76, EVT-P-0001117-T01-IM3-100, EVT-P-0001437-T01-IM3-74, EVT-P-0001840-T01-IM3-185, EVT-P-0001840-T02-IM5-186, ... (40 total)
1705	1755	50	20	98	19	88	2.159090909090909	0.6875140235205702	0.16271843891517399	EVT-P-0001117-T01-IM3-100, EVT-P-0001840-T01-IM3-185, EVT-P-0001840-T02-IM5-186, EVT-P-0004414-T01-IM5-190, EVT-P-0004946-T02-IM5-192, ... (20 total)
1725	1750	25	19	99	18	89	2.0224719101123596	1.0	-0.0	EVT-P-0001840-T01-IM3-185, EVT-P-0001840-T02-IM5-186, EVT-P-0004414-T01-IM5-190, EVT-P-0004946-T02-IM5-192, EVT-P-0012008-T01-IM5-198, ... (19 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
1726	1926	200	106	12	98	9	4.083333333333333	0.08346243364664452	1.0785089560261871	EVT-P-0000569-T01-IM3-76, EVT-P-0000632-T01-IM3-70, EVT-P-0000840-T01-IM3-78, EVT-P-0000840-T03-IM6-22, EVT-P-0001117-T01-IM3-100, ... (106 total)
1730	1755	25	20	98	19	88	2.159090909090909	0.6875140235205702	0.16271843891517399	EVT-P-0001117-T01-IM3-100, EVT-P-0001840-T01-IM3-185, EVT-P-0001840-T02-IM5-186, EVT-P-0004414-T01-IM5-190, EVT-P-0004946-T02-IM5-192, ... (20 total)
1739	1789	50	39	79	36	71	1.352112676056338	1.0	-0.0	EVT-P-0000569-T01-IM3-76, EVT-P-0001117-T01-IM3-100, EVT-P-0001437-T01-IM3-74, EVT-P-0001840-T01-IM3-185, EVT-P-0001840-T02-IM5-186, ... (39 total)
1740	1840	100	42	76	39	68	1.5294117647058822	0.7442597575635433	0.12827546285762348	EVT-P-0000569-T01-IM3-76, EVT-P-0001117-T01-IM3-100, EVT-P-0001437-T01-IM3-74, EVT-P-0001840-T01-IM3-185, EVT-P-0001840-T02-IM5-186, ... (42 total)
1750	1775	25	20	98	19	88	2.159090909090909	0.6875140235205702	0.16271843891517399	EVT-P-0001117-T01-IM3-100, EVT-P-0001840-T01-IM3-185, EVT-P-0001840-T02-IM5-186, EVT-P-0004414-T01-IM5-190, EVT-P-0004946-T02-IM5-192, ... (20 total)
1750	1800	50	40	78	37	70	1.4095238095238096	0.7476634359140739	0.12629385777653632	EVT-P-0000569-T01-IM3-76, EVT-P-0001117-T01-IM3-100, EVT-P-0001437-T01-IM3-74, EVT-P-0001840-T01-IM3-185, EVT-P-0001840-T02-IM5-186, ... (40 total)
1750	1850	100	42	76	39	68	1.5294117647058822	0.7442597575635433	0.12827546285762348	EVT-P-0000569-T01-IM3-76, EVT-P-0001117-T01-IM3-100, EVT-P-0001437-T01-IM3-74, EVT-P-0001840-T01-IM3-185, EVT-P-0001840-T02-IM5-186, ... (42 total)
1750	1950	200	106	12	98	9	4.083333333333333	0.08346243364664452	1.0785089560261871	EVT-P-0000569-T01-IM3-76, EVT-P-0000632-T01-IM3-70, EVT-P-0000840-T01-IM3-78, EVT-P-0000840-T03-IM6-22, EVT-P-0001117-T01-IM3-100, ... (106 total)
1753	1853	100	71	47	65	42	1.2896825396825398	0.7520920961097244	0.12372897538802882	EVT-P-0000569-T01-IM3-76, EVT-P-0000632-T01-IM3-70, EVT-P-0000840-T01-IM3-78, EVT-P-0000840-T03-IM6-22, EVT-P-0001117-T01-IM3-100, ... (71 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
1755	1805	50	21	97	19	88	0.9715909090909091	1.0	-0.0	EVT-P-0000569-T01-IM3-76, EVT-P-0001117-T01-IM3-100, EVT-P-0001437-T01-IM3-74, EVT-P-0003970-T01-IM5-103, EVT-P-0003970-T02-IM5-34, ... (21 total)
1755	1855	100	71	47	65	42	1.2896825396825398	0.7520920961097244	0.12372897538802882	EVT-P-0000569-T01-IM3-76, EVT-P-0000632-T01-IM3-70, EVT-P-0000840-T01-IM3-78, EVT-P-0000840-T03-IM6-22, EVT-P-0001117-T01-IM3-100, ... (71 total)
1755	1955	200	87	31	80	27	1.693121693121693	0.4762239226040929	0.3221887920600883	EVT-P-0000569-T01-IM3-76, EVT-P-0000632-T01-IM3-70, EVT-P-0000840-T01-IM3-78, EVT-P-0000840-T03-IM6-22, EVT-P-0001117-T01-IM3-100, ... (87 total)
1764	1789	25	19	99	17	90	0.85	1.0	-0.0	EVT-P-0000569-T01-IM3-76, EVT-P-0001437-T01-IM3-74, EVT-P-0003970-T01-IM5-103, EVT-P-0003970-T02-IM5-34, EVT-P-0004747-T01-IM5-67, ... (19 total)
1775	1800	25	20	98	18	89	0.9101123595505618	1.0	-0.0	EVT-P-0000569-T01-IM3-76, EVT-P-0001437-T01-IM3-74, EVT-P-0003970-T01-IM5-103, EVT-P-0003970-T02-IM5-34, EVT-P-0004747-T01-IM5-67, ... (20 total)
1781	1881	100	84	34	77	30	1.4666666666666666	0.7273569058478362	0.13825243342552176	EVT-P-0000569-T01-IM3-76, EVT-P-0000632-T01-IM3-70, EVT-P-0000840-T01-IM3-78, EVT-P-0000840-T03-IM6-22, EVT-P-0001437-T01-IM3-74, ... (84 total)
1789	1814	25	20	98	18	89	0.9101123595505618	1.0	-0.0	EVT-P-0000569-T01-IM3-76, EVT-P-0001437-T01-IM3-74, EVT-P-0003970-T01-IM5-103, EVT-P-0003970-T02-IM5-34, EVT-P-0004747-T01-IM5-67, ... (20 total)
1789	1839	50	20	98	18	89	0.9101123595505618	1.0	-0.0	EVT-P-0000569-T01-IM3-76, EVT-P-0001437-T01-IM3-74, EVT-P-0003970-T01-IM5-103, EVT-P-0003970-T02-IM5-34, EVT-P-0004747-T01-IM5-67, ... (20 total)
1789	1889	100	84	34	77	30	1.4666666666666666	0.7273569058478362	0.13825243342552176	EVT-P-0000569-T01-IM3-76, EVT-P-0000632-T01-IM3-70, EVT-P-0000840-T01-IM3-78, EVT-P-0000840-T03-IM6-22, EVT-P-0001437-T01-IM3-74, ... (84 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
1789	1989	200	86	32	79	28	1.6122448979591837	0.48656743023969135	0.3128569651556764	EVT-P-0000569-T01-IM3-76, EVT-P-0000632-T01-IM3-70, EVT-P-0000840-T01-IM3-78, EVT-P-0000840-T03-IM6-22, EVT-P-0001437-T01-IM3-74, ... (86 total)
1800	1900	100	65	53	60	47	1.5319148936170213	0.5392135184056009	0.2682392283136831	EVT-P-0000632-T01-IM3-70, EVT-P-0000840-T01-IM3-78, EVT-P-0000840-T03-IM6-22, EVT-P-0002707-T01-IM3-66, EVT-P-0002707-T03-IM5-59, ... (65 total)
1800	2000	200	67	51	62	45	1.6533333333333333	0.5278673165278271	0.27747522697617966	EVT-P-0000632-T01-IM3-70, EVT-P-0000840-T01-IM3-78, EVT-P-0000840-T03-IM6-22, EVT-P-0002707-T01-IM3-66, EVT-P-0002707-T03-IM5-59, ... (67 total)
1803	1853	50	50	68	46	61	1.319672131147541	0.7576318295964224	0.1205417880600774	EVT-P-0000632-T01-IM3-70, EVT-P-0000840-T01-IM3-78, EVT-P-0000840-T03-IM6-22, EVT-P-0002707-T01-IM3-66, EVT-P-0002707-T03-IM5-59, ... (50 total)
1826	1926	100	66	52	61	46	1.5913043478260869	0.532857842797402	0.27338863773373334	EVT-P-0000632-T01-IM3-70, EVT-P-0000840-T01-IM3-78, EVT-P-0000840-T03-IM6-22, EVT-P-0002707-T01-IM3-66, EVT-P-0002707-T03-IM5-59, ... (66 total)
1828	1853	25	50	68	46	61	1.319672131147541	0.7576318295964224	0.1205417880600774	EVT-P-0000632-T01-IM3-70, EVT-P-0000840-T01-IM3-78, EVT-P-0000840-T03-IM6-22, EVT-P-0002707-T01-IM3-66, EVT-P-0002707-T03-IM5-59, ... (50 total)
1831	1881	50	64	54	59	48	1.475	0.7521953192062142	0.12366937345182379	EVT-P-0000632-T01-IM3-70, EVT-P-0000840-T01-IM3-78, EVT-P-0000840-T03-IM6-22, EVT-P-0002707-T01-IM3-66, EVT-P-0002707-T03-IM5-59, ... (64 total)
1840	1865	25	50	68	46	61	1.319672131147541	0.7576318295964224	0.1205417880600774	EVT-P-0000632-T01-IM3-70, EVT-P-0000840-T01-IM3-78, EVT-P-0000840-T03-IM6-22, EVT-P-0002707-T01-IM3-66, EVT-P-0002707-T03-IM5-59, ... (50 total)
1840	1890	50	64	54	59	48	1.475	0.7521953192062142	0.12366937345182379	EVT-P-0000632-T01-IM3-70, EVT-P-0000840-T01-IM3-78, EVT-P-0000840-T03-IM6-22, EVT-P-0002707-T01-IM3-66, EVT-P-0002707-T03-IM5-59, ... (64 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
1840	1940	100	66	52	61	46	1.5913043478260869	0.532857842797402	0.27338863773373334	EVT-P-0000632-T01-IM3-70, EVT-P-0000840-T01-IM3-78, EVT-P-0000840-T03-IM6-22, EVT-P-0002707-T01-IM3-66, EVT-P-0002707-T03-IM5-59, ... (66 total)
1840	2040	200	66	52	61	46	1.5913043478260869	0.532857842797402	0.27338863773373334	EVT-P-0000632-T01-IM3-70, EVT-P-0000840-T01-IM3-78, EVT-P-0000840-T03-IM6-22, EVT-P-0002707-T01-IM3-66, EVT-P-0002707-T03-IM5-59, ... (66 total)
1853	1878	25	48	70	44	63	1.2222222222222223	1.0	-0.0	EVT-P-0000632-T01-IM3-70, EVT-P-0000840-T01-IM3-78, EVT-P-0000840-T03-IM6-22, EVT-P-0002707-T01-IM3-66, EVT-P-0002707-T03-IM5-59, ... (48 total)
1853	1903	50	62	56	57	50	1.368	0.7545842859001414	0.12229224315558548	EVT-P-0000632-T01-IM3-70, EVT-P-0000840-T01-IM3-78, EVT-P-0000840-T03-IM6-22, EVT-P-0002707-T01-IM3-66, EVT-P-0002707-T03-IM5-59, ... (62 total)
1853	1953	100	64	54	59	48	1.475	0.7521953192062142	0.12366937345182379	EVT-P-0000632-T01-IM3-70, EVT-P-0000840-T01-IM3-78, EVT-P-0000840-T03-IM6-22, EVT-P-0002707-T01-IM3-66, EVT-P-0002707-T03-IM5-59, ... (64 total)
1853	2053	200	64	54	59	48	1.475	0.7521953192062142	0.12366937345182379	EVT-P-0000632-T01-IM3-70, EVT-P-0000840-T01-IM3-78, EVT-P-0000840-T03-IM6-22, EVT-P-0002707-T01-IM3-66, EVT-P-0002707-T03-IM5-59, ... (64 total)
1856	1881	25	14	104	13	94	1.3829787234042554	1.0	-0.0	EVT-P-0003207-T01-IM5-183, EVT-P-0004445-T01-IM5-107, EVT-P-0005086-T01-IM5-108, EVT-P-0007626-T01-IM5-109, EVT-P-0019111-T01-IM6-111, ... (14 total)
1876	1926	50	16	102	15	92	1.6304347826086956	1.0	-0.0	EVT-P-0003207-T01-IM5-183, EVT-P-0004445-T01-IM5-107, EVT-P-0005086-T01-IM5-108, EVT-P-0007626-T01-IM5-109, EVT-P-0019111-T01-IM6-111, ... (16 total)
1881	1906	25	14	104	13	94	1.3829787234042554	1.0	-0.0	EVT-P-0003207-T01-IM5-183, EVT-P-0004445-T01-IM5-107, EVT-P-0005086-T01-IM5-108, EVT-P-0007626-T01-IM5-109, EVT-P-0019111-T01-IM6-111, ... (14 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
1881	1931	50	16	102	15	92	1.6304347826086956	1.0	-0.0	EVT-P-0003207-T01-IM5-183, EVT-P-0004445-T01-IM5-107, EVT-P-0005086-T01-IM5-108, EVT-P-0007626-T01-IM5-109, EVT-P-0019111-T01-IM6-111, ... (16 total)
1881	1981	100	16	102	15	92	1.6304347826086956	1.0	-0.0	EVT-P-0003207-T01-IM5-183, EVT-P-0004445-T01-IM5-107, EVT-P-0005086-T01-IM5-108, EVT-P-0007626-T01-IM5-109, EVT-P-0019111-T01-IM6-111, ... (16 total)
1881	2081	200	16	102	15	92	1.6304347826086956	1.0	-0.0	EVT-P-0003207-T01-IM5-183, EVT-P-0004445-T01-IM5-107, EVT-P-0005086-T01-IM5-108, EVT-P-0007626-T01-IM5-109, EVT-P-0019111-T01-IM6-111, ... (16 total)

top_windows

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
1593	1693	100	4	114	2	105	0.08571428571428572	0.042988935779828465	1.366643306003459	EVT-P-0007870-T02-IM6-207, EVT-P-0037864-T02-IM7-49, EVT-P-0054667-T01-IM6-30, EVT-P-0055135-T03-IM7-90
1694	1894	200	107	11	99	8	4.640625	0.06612721119379524	1.1796197925642002	EVT-P-0000569-T01-IM3-76, EVT-P-0000632-T01-IM3-70, EVT-P-0000840-T01-IM3-78, EVT-P-0000840-T03-IM6-22, EVT-P-0001117-T01-IM3-100, ... (107 total)
1693	1718	25	5	113	3	104	0.12980769230769232	0.06794655006722032	1.1678325893333363	EVT-P-0007870-T02-IM6-207, EVT-P-0016046-T01-IM6-171, EVT-P-0037022-T01-IM6-1, EVT-P-0054497-T01-IM6-8, EVT-P-0055135-T03-IM7-90
1726	1926	200	106	12	98	9	4.083333333333333	0.08346243364664452	1.0785089560261871	EVT-P-0000569-T01-IM3-76, EVT-P-0000632-T01-IM3-70, EVT-P-0000840-T01-IM3-78, EVT-P-0000840-T03-IM6-22, EVT-P-0001117-T01-IM3-100, ... (106 total)
1669	1694	25	6	112	4	103	0.17475728155339806	0.09665884968988354	1.0147583775843583	EVT-P-0007870-T02-IM6-207, EVT-P-0016046-T01-IM6-171, EVT-P-0037022-T01-IM6-1, EVT-P-0054497-T01-IM6-8, EVT-P-0054667-T01-IM6-30, ... (6 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
1594	1694	100	7	111	5	102	0.22058823529411764	0.12834492320203683	0.8916213055585486	EVT-P-0007870-T02-IM6-207, EVT-P-0016046-T01-IM6-171, EVT-P-0037022-T01-IM6-1, EVT-P-0037864-T02-IM7-49, EVT-P-0054497-T01-IM6-8, ... (7 total)
1755	1955	200	87	31	80	27	1.693121693121693	0.4762239226040929	0.3221887920600883	EVT-P-0000569-T01-IM3-76, EVT-P-0000632-T01-IM3-70, EVT-P-0000840-T01-IM3-78, EVT-P-0000840-T03-IM6-22, EVT-P-0001117-T01-IM3-100, ... (87 total)
1789	1989	200	86	32	79	28	1.6122448979591837	0.48656743023969135	0.3128569651556764	EVT-P-0000569-T01-IM3-76, EVT-P-0000632-T01-IM3-70, EVT-P-0000840-T01-IM3-78, EVT-P-0000840-T03-IM6-22, EVT-P-0001437-T01-IM3-74, ... (86 total)
1800	2000	200	67	51	62	45	1.6533333333333333	0.5278673165278271	0.27747522697617966	EVT-P-0000632-T01-IM3-70, EVT-P-0000840-T01-IM3-78, EVT-P-0000840-T03-IM6-22, EVT-P-0002707-T01-IM3-66, EVT-P-0002707-T03-IM5-59, ... (67 total)
1826	1926	100	66	52	61	46	1.5913043478260869	0.532857842797402	0.27338863773373334	EVT-P-0000632-T01-IM3-70, EVT-P-0000840-T01-IM3-78, EVT-P-0000840-T03-IM6-22, EVT-P-0002707-T01-IM3-66, EVT-P-0002707-T03-IM5-59, ... (66 total)
1800	1900	100	65	53	60	47	1.5319148936170213	0.5392135184056009	0.2682392283136831	EVT-P-0000632-T01-IM3-70, EVT-P-0000840-T01-IM3-78, EVT-P-0000840-T03-IM6-22, EVT-P-0002707-T01-IM3-66, EVT-P-0002707-T03-IM5-59, ... (65 total)
1681	1881	200	110	8	100	7	1.4285714285714286	0.5544227008978931	0.25615899579731366	EVT-P-0000569-T01-IM3-76, EVT-P-0000632-T01-IM3-70, EVT-P-0000840-T01-IM3-78, EVT-P-0000840-T03-IM6-22, EVT-P-0001117-T01-IM3-100, ... (110 total)
1730	1755	25	20	98	19	88	2.159090909090909	0.6875140235205702	0.16271843891517399	EVT-P-0001117-T01-IM3-100, EVT-P-0001840-T01-IM3-185, EVT-P-0001840-T02-IM5-186, EVT-P-0004414-T01-IM5-190, EVT-P-0004946-T02-IM5-192, ... (20 total)
1650	1750	100	25	93	22	85	0.6901960784313725	0.698430794091426	0.15587662066395394	EVT-P-0001840-T01-IM3-185, EVT-P-0001840-T02-IM5-186, EVT-P-0004414-T01-IM5-190, EVT-P-0004946-T02-IM5-192, EVT-P-0007870-T02-IM6-207, ... (25 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
1655	1755	100	26	92	23	84	0.7301587301587301	0.7049161533582773	0.1518625373339728	EVT-P-0001117-T01-IM3-100, EVT-P-0001840-T01-IM3-185, EVT-P-0001840-T02-IM5-186, EVT-P-0004414-T01-IM5-190, EVT-P-0004946-T02-IM5-192, ... (26 total)
1550	1750	200	26	92	23	84	0.7301587301587301	0.7049161533582773	0.1518625373339728	EVT-P-0001840-T01-IM3-185, EVT-P-0001840-T02-IM5-186, EVT-P-0004414-T01-IM5-190, EVT-P-0004946-T02-IM5-192, EVT-P-0007870-T02-IM6-207, ... (26 total)
1555	1755	200	27	91	24	83	0.7710843373493976	0.7124856955112606	0.14722385050947034	EVT-P-0001117-T01-IM3-100, EVT-P-0001840-T01-IM3-185, EVT-P-0001840-T02-IM5-186, EVT-P-0004414-T01-IM5-190, EVT-P-0004946-T02-IM5-192, ... (27 total)
1781	1881	100	84	34	77	30	1.4666666666666666	0.7273569058478362	0.13825243342552176	EVT-P-0000569-T01-IM3-76, EVT-P-0000632-T01-IM3-70, EVT-P-0000840-T01-IM3-78, EVT-P-0000840-T03-IM6-22, EVT-P-0001437-T01-IM3-74, ... (84 total)
1694	1794	100	42	76	39	68	1.5294117647058822	0.7442597575635433	0.12827546285762348	EVT-P-0000569-T01-IM3-76, EVT-P-0001117-T01-IM3-100, EVT-P-0001437-T01-IM3-74, EVT-P-0001840-T01-IM3-185, EVT-P-0001840-T02-IM5-186, ... (42 total)
1740	1840	100	42	76	39	68	1.5294117647058822	0.7442597575635433	0.12827546285762348	EVT-P-0000569-T01-IM3-76, EVT-P-0001117-T01-IM3-100, EVT-P-0001437-T01-IM3-74, EVT-P-0001840-T01-IM3-185, EVT-P-0001840-T02-IM5-186, ... (42 total)
1689	1789	100	44	74	39	68	0.6882352941176471	0.7446835754012113	0.12802822476719072	EVT-P-0000569-T01-IM3-76, EVT-P-0001117-T01-IM3-100, EVT-P-0001437-T01-IM3-74, EVT-P-0001840-T01-IM3-185, EVT-P-0001840-T02-IM5-186, ... (44 total)
1750	1800	50	40	78	37	70	1.4095238095238096	0.7476634359140739	0.12629385777653632	EVT-P-0000569-T01-IM3-76, EVT-P-0001117-T01-IM3-100, EVT-P-0001437-T01-IM3-74, EVT-P-0001840-T01-IM3-185, EVT-P-0001840-T02-IM5-186, ... (40 total)
1589	1789	200	46	72	41	66	0.7454545454545455	0.7487650178578418	0.1256544539416864	EVT-P-0000569-T01-IM3-76, EVT-P-0001117-T01-IM3-100, EVT-P-0001437-T01-IM3-74, EVT-P-0001840-T01-IM3-185, EVT-P-0001840-T02-IM5-186, ... (46 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
1753	1853	100	71	47	65	42	1.2896825396825398	0.7520920961097244	0.12372897538802882	EVT-P-0000569-T01-IM3-76, EVT-P-0000632-T01-IM3-70, EVT-P-0000840-T01-IM3-78, EVT-P-0000840-T03-IM6-22, EVT-P-0001117-T01-IM3-100, ... (71 total)
1600	1800	200	47	71	42	65	0.7753846153846153	0.7520920961097245	0.12372897538802877	EVT-P-0000569-T01-IM3-76, EVT-P-0001117-T01-IM3-100, EVT-P-0001437-T01-IM3-74, EVT-P-0001840-T01-IM3-185, EVT-P-0001840-T02-IM5-186, ... (47 total)
1831	1881	50	64	54	59	48	1.475	0.7521953192062142	0.12366937345182379	EVT-P-0000632-T01-IM3-70, EVT-P-0000840-T01-IM3-78, EVT-P-0000840-T03-IM6-22, EVT-P-0002707-T01-IM3-66, EVT-P-0002707-T03-IM5-59, ... (64 total)
1853	1953	100	64	54	59	48	1.475	0.7521953192062142	0.12366937345182379	EVT-P-0000632-T01-IM3-70, EVT-P-0000840-T01-IM3-78, EVT-P-0000840-T03-IM6-22, EVT-P-0002707-T01-IM3-66, EVT-P-0002707-T03-IM5-59, ... (64 total)
1853	1903	50	62	56	57	50	1.368	0.7545842859001414	0.12229224315558548	EVT-P-0000632-T01-IM3-70, EVT-P-0000840-T01-IM3-78, EVT-P-0000840-T03-IM6-22, EVT-P-0002707-T01-IM3-66, EVT-P-0002707-T03-IM5-59, ... (62 total)
1640	1840	200	48	70	43	64	0.80625	0.7562186041130197	0.12135264254576777	EVT-P-0000569-T01-IM3-76, EVT-P-0001117-T01-IM3-100, EVT-P-0001437-T01-IM3-74, EVT-P-0001840-T01-IM3-185, EVT-P-0001840-T02-IM5-186, ... (48 total)
1828	1853	25	50	68	46	61	1.319672131147541	0.7576318295964224	0.1205417880600774	EVT-P-0000632-T01-IM3-70, EVT-P-0000840-T01-IM3-78, EVT-P-0000840-T03-IM6-22, EVT-P-0002707-T01-IM3-66, EVT-P-0002707-T03-IM5-59, ... (50 total)
1725	1750	25	19	99	18	89	2.0224719101123596	1.0	-0.0	EVT-P-0001840-T01-IM3-185, EVT-P-0001840-T02-IM5-186, EVT-P-0004414-T01-IM5-190, EVT-P-0004946-T02-IM5-192, EVT-P-0012008-T01-IM5-198, ... (19 total)
1764	1789	25	19	99	17	90	0.85	1.0	-0.0	EVT-P-0000569-T01-IM3-76, EVT-P-0001437-T01-IM3-74, EVT-P-0003970-T01-IM5-103, EVT-P-0003970-T02-IM5-34, EVT-P-0004747-T01-IM5-67, ... (19 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
1775	1800	25	20	98	18	89	0.9101123595505618	1.0	-0.0	EVT-P-0000569-T01-IM3-76, EVT-P-0001437-T01-IM3-74, EVT-P-0003970-T01-IM5-103, EVT-P-0003970-T02-IM5-34, EVT-P-0004747-T01-IM5-67, ... (20 total)
1853	1878	25	48	70	44	63	1.2222222222222223	1.0	-0.0	EVT-P-0000632-T01-IM3-70, EVT-P-0000840-T01-IM3-78, EVT-P-0000840-T03-IM6-22, EVT-P-0002707-T01-IM3-66, EVT-P-0002707-T03-IM5-59, ... (48 total)
1856	1881	25	14	104	13	94	1.3829787234042554	1.0	-0.0	EVT-P-0003207-T01-IM5-183, EVT-P-0004445-T01-IM5-107, EVT-P-0005086-T01-IM5-108, EVT-P-0007626-T01-IM5-109, EVT-P-0019111-T01-IM6-111, ... (14 total)
1739	1789	50	39	79	36	71	1.352112676056338	1.0	-0.0	EVT-P-0000569-T01-IM3-76, EVT-P-0001117-T01-IM3-100, EVT-P-0001437-T01-IM3-74, EVT-P-0001840-T01-IM3-185, EVT-P-0001840-T02-IM5-186, ... (39 total)
1755	1805	50	21	97	19	88	0.9715909090909091	1.0	-0.0	EVT-P-0000569-T01-IM3-76, EVT-P-0001117-T01-IM3-100, EVT-P-0001437-T01-IM3-74, EVT-P-0003970-T01-IM5-103, EVT-P-0003970-T02-IM5-34, ... (21 total)
1876	1926	50	16	102	15	92	1.6304347826086956	1.0	-0.0	EVT-P-0003207-T01-IM5-183, EVT-P-0004445-T01-IM5-107, EVT-P-0005086-T01-IM5-108, EVT-P-0007626-T01-IM5-109, EVT-P-0019111-T01-IM6-111, ... (16 total)
1653	1853	200	96	22	87	20	0.9666666666666667	1.0	-0.0	EVT-P-0000569-T01-IM3-76, EVT-P-0000632-T01-IM3-70, EVT-P-0000840-T01-IM3-78, EVT-P-0000840-T03-IM6-22, EVT-P-0001117-T01-IM3-100, ... (96 total)
1693	1893	200	109	9	99	8	1.2375	1.0	-0.0	EVT-P-0000569-T01-IM3-76, EVT-P-0000632-T01-IM3-70, EVT-P-0000840-T01-IM3-78, EVT-P-0000840-T03-IM6-22, EVT-P-0001117-T01-IM3-100, ... (109 total)

Per-event results (results.tsv)

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_effect_on_frame
EVT-P-0037022-T01-IM6-1	P-0037022-T01-IM6		Prostate Cancer	PRAD	AFG1L-ROS1	AFG1L	in-frame	three_prime	117660863	31	1694	True	1694	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		LACE1 (NM_145315) - ROS1 (NM_002944) rearrangement: c.808-19179:LACE1_c.5079+1435:ROS1inv Protein Fusion: in frame {LACE1: ROS1}	NA
EVT-P-0041695-T01-IM6-3	P-0041695-T01-IM6		Non-Small Cell Lung Cancer	LUAD	CCDC30-ROS1	CCDC30	out-of-frame	three_prime	117643359	35	1881	True	1881	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		CCDC30 (NM_001080850) - ROS1 (NM_002944) fusion: t(1;6)(p34.2;q22.1)(chr1:g.43087864::chr6:g.117643359) Protein Fusion: out of frame {CCDC30: ROS1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frameshift	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_effect_on_frame
EVT-P-0014506-T02-IM6-6	P-0014506-T02-IM6		Non-Small Cell Lung Cancer	LUAD	CD74-ROS1	CD74	unknown	three_prime	117650623	32	1750	True	1750	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149782872::chr6:g.117650623) Protein Fusion: mid-exon {CD74:ROS1}	NA
EVT-P-0011025-T02-IM6-7	P-0011025-T02-IM6		Non-Small Cell Lung Cancer	LUAD	CD74-ROS1	CD74	in-frame	three_prime	117645889	34	1853	True	1853	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149783756::chr6:g.117645889) Protein Fusion: in frame {CD74:ROS1}	NA

event_id	sample_id	patient_id	tumor_type	oncotre_e_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0054497-T01-IM6-8	P-0054497-T01-IM6		Non-Small Cell Lung Cancer	LUAD	CD74-ROS1	CD74	out-of-frame	three_prime	117659373	31	1694	True	1694	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149783252::chr6:g.117659373) Protein Fusion: out of frame {CD74:ROS1}	NA
EVT-P-0054440-T02-IM6-9	P-0054440-T02-IM6		Non-Small Cell Lung Cancer	LUAD	CD74-ROS1	CD74	in-frame	three_prime	117646088	34	1853	True	1853	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149782403::chr6:g.117646088) Protein Fusion: in frame {CD74:ROS1}	NA

event_id	sample_id	patient_id	tumor_type	oncotre_e_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0054352-T01-IM6-10	P-0054352-T01-I-M6		Non-Small Cell Lung Cancer	LUAD	CD74-ROS1	CD74	out-of-frame	three_prime	117649394	33	1789	True	1789	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149783536::chr6:g.117649394) Protein Fusion: out of frame {CD74:ROS1}	NA
EVT-P-0014506-T05-IM7-11	P-0014506-T05-I-M7		Non-Small Cell Lung Cancer	LUAD	CD74-ROS1	CD74	unknown	three_prime	117650623	32	1750	True	1750	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149782872::chr6:g.117650623) Protein Fusion: mid-exon {CD74:ROS1}	NA

event_id	sample_id	patient_id	tumor_type	oncotre_e_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0026455-T01-IM6-12	P-0026455-T01-IM6		Non-Small Cell Lung Cancer	LUAD	CD74-ROS1	CD74	in-frame	three_prime	117646566	34	1853	True	1853	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-6 fused to ROS1 exons 34-43): t(5;6)(q32;q22.1)(c chr5:g.149783651::chr6:g.117646566) Protein Fusion: in frame {CD74:ROS1}	NA
EVT-P-0052363-T02-IM7-13	P-0052363-T02-IM7		Non-Small Cell Lung Cancer	LUAD	CD74-ROS1	CD74	in-frame	three_prime	117647256	34	1853	True	1853	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(c chr5:g.149783700::chr6:g.117647256) Protein Fusion: in frame {CD74:ROS1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_effect_on_frame
EVT-P-0052363-T01-IM6-14	P-0052363-T01-IM6		Non-Small Cell Lung Cancer	LUAD	CD74-ROS1	CD74	in-frame	three_prime	117647256	34	1853	True	1853	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149783692::chr6:g.117647256) Protein Fusion: in frame {CD74:ROS1}	NA
EVT-P-0049640-T03-IM7-15	P-0049640-T03-IM7		Non-Small Cell Lung Cancer	LUAD	CD74-ROS1	CD74	out-of-frame	three_prime	117650417	33	1789	True	1789	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149782214::chr6:g.117650417) Protein Fusion: out of frame {CD74:ROS1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0049640-T01-IM6-16	P-0049640-T01-IM6		Non-Small Cell Lung Cancer	LUAD	CD74-ROS1	CD74	out-of-frame	three_prime	117650417	33	1789	True	1789	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149782214::chr6:g.117650417) Protein Fusion: out of frame {CD74:ROS1}	NA
EVT-P-0048671-T02-IM6-17	P-0048671-T02-IM6		Non-Small Cell Lung Cancer	LUAD	CD74-ROS1	CD74	in-frame	three_prime	117647250	34	1853	True	1853	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149783869::chr6:g.117647250) Protein Fusion: in frame {CD74:ROS1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0011025-T01-IM5-18	P-0011025-T01-IM5		Non-Small Cell Lung Cancer	LUAD	CD74-ROS1	CD74	in-frame	three_prime	117645889	34	1853	True	1853	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-6 fused with ROS1 exons 34-43) : t(5;6)(q32;q22.1)(c14hr5:g.149783755::chr6:g.117645889) Protein fusion: in frame (CD74-ROS1)	NA
EVT-P-0048671-T01-IM6-19	P-0048671-T01-IM6		Non-Small Cell Lung Cancer	LUAD	CD74-ROS1	CD74	in-frame	three_prime	117647250	34	1853	True	1853	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(c14hr5:g.149783869::chr6:g.117647250) Protein Fusion: in frame {CD74:ROS1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_effect_on_frame
EVT-P-0047502-T01-IM6-20	P-0047502-T01-IM6		Non-Small Cell Lung Cancer	LUAS	CD74-ROS1	CD74	in-frame	three_prime	117646405	34	1853	True	1853	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149782474::chr6:g.117646405) Protein Fusion: in frame {CD74:ROS1}	NA
EVT-P-004465-T01-IM6-21	P-004465-T01-IM6		Non-Small Cell Lung Cancer	LUAD	CD74-ROS1	CD74	in-frame	three_prime	117646909	34	1853	True	1853	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149783638::chr6:g.117646909) Protein Fusion: in frame {CD74:ROS1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_effect_on_frame
EVT-P-000840-T03-IM6-22	P-0000840-T03-IM6		Non-Small Cell Lung Cancer	LUAD	CD74-ROS1	CD74	in-frame	three_prime	117646938	34	1853	True	1853	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149783509::chr6:g.117646938) Protein Fusion: in frame {CD74:ROS1}	NA
EVT-P-004118-T06-IM6-23	P-004118-T06-IM6		Non-Small Cell Lung Cancer	LUAD	CD74-ROS1	CD74	unknown	three_prime	117647425	33	1840	False	1840	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149782265::chr6:g.117647425) Protein Fusion: mid-exon {CD74:ROS1}	NA

event_id	sample_id	patient_id	tumor_type	oncotre_e_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0041118-T02-IM6-24	P-0041118-T02-IM6		Non-Small Cell Lung Cancer	LUAD	CD74-ROS1	CD74	unknown	three_prime	117647424	33	1840	False	1840	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149782265::chr6:g.117647424) Protein Fusion: mid-exon {CD74:ROS1}	NA
EVT-P-0004747-T04-IM7-25	P-0004747-T04-IM7		Non-Small Cell Lung Cancer	LUAD	CD74-ROS1	CD74	out-of-frame	three_prime	117649107	33	1789	True	1789	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149782562::chr6:g.117649107) Protein Fusion: out of frame {CD74:ROS1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0040869-T01-IM6-26	P-0040869-T01-IM6		Non-Small Cell Lung Cancer	LUAD	CD74-ROS1	CD74	in-frame	three_prime	117647263	34	1853	True	1853	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149783792::chr6:g.117647263) Protein Fusion: in frame {CD74:ROS1}	NA
EVT-P-0040577-T01-IM6-27	P-0040577-T01-IM6		Non-Small Cell Lung Cancer	LUAD	CD74-ROS1	CD74	out-of-frame	three_prime	117650159	33	1789	True	1789	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-6 fused with ROS1 exons 32-43) : t(5;6)(q32;q22.1)(chr5:g.149783441::chr6:g.117650159) Protein Fusion: out of frame {CD74:ROS1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0039219-T01-IM6-28	P-0039219-T01-IM6		Non-Small Cell Lung Cancer	LUAD	CD74-ROS1	CD74	in-frame	three_prime	117646339	34	1853	True	1853	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q32;q22.1)(chr5:g.149783535::chr6:g.117646339) Protein Fusion: in frame {CD74:ROS1}	NA
EVT-P-0024572-T01-IM6-29	P-0024572-T01-IM6		Non-Small Cell Lung Cancer	LUAD	CD74-ROS1	CD74	out-of-frame	three_prime	117649758	33	1789	True	1789	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion(CD74 exons 1-6 fused to ROS1 exons 33-43): t(5;6)(q32;q22.1)(chr5:g.149783583::chr6:g.117649758) Protein Fusion: out of frame {CD74:ROS1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0054667-T01-IM6-30	P-0054667-T01-IM6		Ampullary Cancer	MAMPCA	CEP85L-ROS1	CEP85L	unknown	three_prime	117662314	30	1688	False	1688	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		CEP85L (NM_001178035) - ROS1 (NM_002944) fusion: c. 1149-2421:CEP85L_c.5063:ROS1del Protein Fusion: mid-exon {CEP85L:ROS1}	NA
EVT-P-0063351-T02-IM7-33	P-0063351-T02-IM7		Non-Small Cell Lung Cancer	LUAD	EZR-ROS1	EZR	in-frame	three_prime	117645650	34	1853	True	1853	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		EZR (NM_003379) - ROS1 (NM_002944) fusion: c. 1091-221:EZR_c.5558-72:ROS1del Protein Fusion: in frame {EZR:ROS1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0003970-T02-IM5-34	P-0003970-T02-IM5		Non-Small Cell Lung Cancer	LUAD	EZR-ROS1	EZR	out-of-frame	three_prime	117649698	33	1789	True	1789	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		EZR (NM_003379) - ROS1 (NM_002944) fusion: (EZR exons 1 to 9 fused ROS1 exons 33 to 43): c. 1090+142:EZR_c.5366_794:ROS1del Protein fusion: out of frame (EZR-ROS1)	NA
EVT-P-0020458-T04-IM6-35	P-0020458-T04-IM6		Non-Small Cell Lung Cancer	LUAD	EZR-ROS1	EZR	in-frame	three_prime	117646033	34	1853	True	1853	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		EZR (NM_003379) - ROS1 (NM_002944) fusion: c. 1090+208:EZR_c.5558-455:ROS1del Protein Fusion: in frame {EZR:ROS1}	NA

event_id	sample_id	patient_id	tumor_type	oncotre_e_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intro_nic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0028081-T01-IM6-36	P-0028081-T01-IM6		Non-Small Cell Lung Cancer	LUAD	EZR-ROS1	EZR	in-frame	three_prime	117647275	34	1853	True	1853	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		EZR (NM_003379) - ROS1 (NM_002944) fusion: c.1090+285:EZR_c.5557+112:ROS1del Protein Fusion: in frame {EZR:ROS1}	NA
EVT-P-0068494-T01-IM7-38	P-0068494-T01-IM7		Soft Tissue Sarcoma	IMT	FN1-ROS1	FN1	in-frame	three_prime	117652028	32	1750	True	1750	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		FN1 (NM_212482) - ROS1 (NM_002944) fusion: t(2;6)(q35;q22.1)(c11hr2:g.216267900::chr6:g.117652028) Protein Fusion: in frame {FN1:ROS1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0021147-T01-IM6-40	P-0021147-T01-IM6		Colorectal Cancer	COAD	GOPC-R OS1	GOPC	in-frame	three_prime	117641701	36	1926	True	1926	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		GOPC (NM_020399) - ROS1 (NM_002944) fusion (GOPC exons 1-4 fused with ROS1 exons 36-43): c.651-735:GOPC_c.5778-508:ROS1del Protein Fusion: in frame {GOPC: ROS1}	NA
EVT-P-0015494-T01-IM6-44	P-0015494-T01-IM6		Breast Cancer	IDC	IGF1R-R OS1	IGF1R	in-frame	three_prime	117646453	34	1853	True	1853	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		IGF1R (NM_000875) - ROS1 (NM_002944) Rearrangement: t(6;15)(q22.1;q26.3)(c11p11.1) chr15:g.7646453-7646453::chr15:g.99477638 Protein fusion: in frame (IGF1R-R OS1)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0068483-T01-IM7-45	P-0068483-T01-IM7		Non-Small Cell Lung Cancer	LUAD	LRIG3-ROS1	LRIG3	in-frame	three_prime	117644742	35	1881	True	1881	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		LRIG3 (NM_153377) - ROS1 (NM_002944) fusion: t(6;12)(q22.1;q14.1)(chr6:g.117644742::chr12:g.59269495) Protein Fusion: in frame {LRIG3: ROS1}	NA
EVT-P-0023007-T01-IM6-46	P-0023007-T01-IM6		Non-Small Cell Lung Cancer	LUAD	LRIG3-ROS1	LRIG3	in-frame	three_prime	117645256	35	1881	True	1881	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		LRIG3 (NM_153377) - ROS1 (NM_002944) fusion: (LRIG3 exons 1-15 fused to ROS1 exons 35 - 43): t(6;12)(q22.1;q14.1)(chr6:g.117645256::chr12:g.59270940) Protein Fusion: in frame {LRIG3: ROS1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0028295-T01-IM6-47	P-0028295-T01-IM6		Non-Small Cell Lung Cancer	LUAD	MYH9-ROS1	MYH9	in-frame	three_prime	117641564	36	1926	True	1926	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		MYH9 (NM_002473) - ROS1 (NM_002944) rearrangement: t(6;22)(q22.1;q12.3)(chr6:g.117641564::chr22:g.36691922) Protein Fusion: in frame {MYH9:ROS1}	NA
EVT-P-0037864-T02-IM7-49	P-0037864-T02-IM7		Nerve Sheath Tumor	MPNST	NKAIN2-ROS1	NKAIN2	unknown	three_prime	117662563	29	1634	False	1634	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		NKAIN2 (NM_001040214) - ROS1 (NM_002944) fusion: c.55-122327: NKAIN2_c.4902: ROS1inv Protein Fusion: mid-exon {NKAIN2:ROS1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0054050-T03-IM6-50	P-0054050-T03-IM6		Glioma	GB	NUDCD3-ROS1	NUDCD3	out-of-frame	three_prime	117645481	35	1881	True	1881	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		NUDCD3 (NM_015332) - ROS1 (NM_002944) fusion: t(6;7)(q22.1;p13)(c11hr6:g.117645481::chr7:g.44523353) Protein Fusion: out of frame {NUDCD3:ROS1}	NA
EVT-P-0054050-T05-IM7-51	P-0054050-T05-IM7		Glioma	GB	NUDCD3-ROS1	NUDCD3	out-of-frame	three_prime	117645481	35	1881	True	1881	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		NUDCD3 (NM_015332) - ROS1 (NM_002944) fusion: t(6;7)(q22.1;p13)(c11hr6:g.117645481::chr7:g.44523353) Protein Fusion: out of frame {NUDCD3:ROS1}	NA
EVT-P-0067287-T01-IM7-52	P-0067287-T01-IM7		Hepatobiliary Cancer	GBAD	PRKN-ROS1	PRKN	unknown	three_prime	117674184	26	1430	False	1430	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		PARK2 (NM_004562) - ROS1 (NM_002944) fusion: c.171+704:PARK2_c.4290:ROS1del Protein Fusion: mid-exon {PARK2:ROS1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0019205-T02-IM6-53	P-0019205-T02-IM6		Non-Small Cell Lung Cancer	LUAD	RBPJL-ROS1	RBPJL	unknown	five_prime	117654098	31	1750	True	1749	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False	Fibronectin type III domain	Protein kinase domain		ROS1 (NM_002944) rearrangement: t(6;20)(q22.1;q13.12)(c11hr6:g.117654098::chr20:g.43949086) Anti sense Fusion	NA
EVT-P-0021228-T01-IM6-57	P-0021228-T01-IM6		Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	CD74	in-frame	three_prime	117646295	34	1853	True	1853	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-6 fused to ROS1 exons 34-43): t(5;6)(q32;q22.1)(chr5:g.149783729::chr6:g.117646295) Protein Fusion: in frame {CD74:ROS1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0010760-T01-IM5-58	P-0010760-T01-IM5		Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	CD74	in-frame	three_prime	117647234	34	1853	True	1853	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-6 fused to ROS1 exons 34-43): t(5;6)(5q32;q22.1)(chr5:g.149783763::chr6:g.117647234) Protein fusion: in frame (CD74-ROS1)	NA
EVT-P-0002707-T03-IM5-59	P-0002707-T03-IM5		Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	CD74	in-frame	three_prime	117646946	34	1853	True	1853	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD274 exons 1-6 fused to ROS1 exons 34-43): t(5;6)(q32;q22.1)(chr5:g.49783748::chr6:g.117646946) Protein fusion: in frame (CD74-ROS1)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0015215-T01-IM6-60	P-0015215-T01-IM6		Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	CD74	out-of-frame	three_prime	117650370	33	1789	True	1789	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-6 fused to ROS1 exons 32-43): t(5;6)(q32;q22.1)(chr5:g.149783057::chr6:g.117650370) Protein fusion: out of frame (CD74-ROS1)	NA
EVT-P-0002877-T01-IM3-61	P-0002877-T01-IM3		Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	CD74	in-frame	three_prime	117645870	34	1853	True	1853	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		NA Protein fusion: in frame (CD74-ROS1)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0012525-T01-IM5-62	P-0012525-T01-IM5		Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	CD74	in-frame	three_prime	117647385	34	1853	True	1853	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-6 with ROS1 exons 34-43): t(5;6)(q32;q22.1)(c chr5:g.149783203::chr6:g.117647385) Protein fusion: in frame (CD74-ROS1)	NA
EVT-P-002912-T01-IM3-63	P-0002912-T01-IM3		Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	CD74	in-frame	five_prime	117646326	33	1853	True	1852	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False	Fibronectin type III domain	Protein kinase domain		NA Protein fusion: in frame (ROS1-CD74)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0004336-T01-IM5-64	P-0004336-T01-IM5		Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	CD74	in-frame	three_prime	117647249	34	1853	True	1853	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-6 with ROS1 exons 34-43) : t(5;6)(5q32;q22.1)(chr5:g.149783071::chr6:g.117647249) Protein fusion: in frame (CD74-ROS1)	NA
EVT-P-0016213-T01-IM6-65	P-0016213-T01-IM6		Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	CD74	out-of-frame	three_prime	117649786	33	1789	True	1789	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-7 fused to ROS1 exons 33-43): t(5;6)(q32;q22.1)(chr5:g.149783418::chr6:g.117649786) Protein Fusion: out of frame (CD74:ROS1)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0002707-T01-IM3-66	P-0002707-T01-I M3		Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	CD74	in-frame	five_prime	117646946	33	1853	True	1852	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False	Fibronectin type III domain	Protein kinase domain		NA Protein fusion: in frame (ROS1-CD74)	NA
EVT-P-0004747-T01-IM5-67	P-0004747-T01-I M5		Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	CD74	out-of-frame	three_prime	117649107	33	1789	True	1789	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-7 fused to ROS1 exons 33-43): t(5;6)(q32;q22.1) (chr5:g.149782561::chr6:g.117649107) Protein fusion: out of frame (CD74-ROS1)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0004790-T01-IM5-68	P-0004790-T01-I M5		Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	CD74	in-frame	three_prime	117646467	34	1853	True	1853	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion (exons 1-7 of CD74 fused to exons 34-43 of ROS1) : t(5;6)(q32;q22.1)(chr5:g.149782207::chr6:g.117646467) Protein fusion: in frame (CD74-ROS1)	NA
EVT-P-0036970-T01-IM6-69	P-0036970-T01-I M6		Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	CD74	out-of-frame	three_prime	117650273	33	1789	True	1789	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion: t(5;6)(q33.1;q22.1)(chr5:g.149783480::chr6:g.117650273) Protein Fusion: out of frame {CD74:ROS1}	NA
EVT-P-000632-T01-IM3-70	P-000632-T01-I M3		Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	CD74	in-frame	three_prime	117646313	34	1853	True	1853	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		NA Protein fusion: in frame (CD74-ROS1)	NA

event_id	sample_id	patient_id	tumor_type	oncotre_e_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0032722-T01-IM6-71	P-0032722-T01-IM6		Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	CD74	unknown	three_prime	117645677	34	1853	True	1853	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-7 fused to ROS1 exons 34-43): t(5;6)(q32;q22.1)(c chr5:g.149782874::chr6:g.117645677) Protein Fusion: mid-exon {CD74: ROS1}	NA
EVT-P-0014506-T01-IM6-72	P-0014506-T01-IM6		Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	CD74	unknown	three_prime	117650623	32	1750	True	1750	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-7 fused to ROS1 exons 32-43): t(5;6)(q32;q22.1)(c chr5:g.149782871::chr6:g.117650623) Protein fusion: mid-exon (CD74-ROS1)	NA

event_id	sample_id	patient_id	tumor_type	oncotre_e_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0018222-T01-IM6-73	P-0018222-T01-IM6		Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	CD74	in-frame	three_prime	117646209	34	1853	True	1853	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-6 fused to ROS1 exons 34-43): t(5;6)(q32;q22.1)(c chr5:g.149783915::chr6:g.117646209) Protein Fusion: in frame {CD74:ROS1}	NA
EVT-P-0001437-T01-IM3-74	P-0001437-T01-IM3		Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	CD74	out-of-frame	three_prime	117649412	33	1789	True	1789	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		None Protein fusion: out of frame (CD74-ROS1)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frameshift	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0018513-T01-IM6-75	P-0018513-T01-IM6		Non-Small Cell Lung Cancer	LUSC	ROS1-CD74	CD74	unknown	three_prime	117647544	33	1800	False	1800	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-6 fused with ROS1 exons 33 to 43): t(5;6)(q32;q22.1)(chr5:g.149783804::chr6:g.117647544) Protein Fusion: mid-exon {CD74: ROS1}	NA
EVT-P-0000569-T01-IM3-76	P-0000569-T01-IM3		Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	CD74	out-of-frame	five_prime	117647622	32	1789	True	1788	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False	Fibronectin type III domain	Protein kinase domain		NA Protein fusion: out of frame (ROS1-CD74)	NA

event_id	sample_id	patient_id	tumor_type	oncotre_e_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0025107-T01-IM6-77	P-0025107-T01-IM6		Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	CD74	in-frame	three_prime	117647170	34	1853	True	1853	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-7 fused to ROS1 exons 34-43): t(5;6)(q32;q22.1)(chr5:g.149782204::chr6:g.117647170) Protein Fusion: in frame {CD74:ROS1}	NA
EVT-P-000840-T01-IM3-78	P-000840-T01-IM3		Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	CD74	in-frame	three_prime	117646938	34	1853	True	1853	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		ROS1 (NM_002944) - CD74 (NM_001025158) Translocation: t(6;5)(q22.1;q32)(chr6:g.117646938::chr5:g.149783508) Protein fusion: in frame (CD74-ROS1)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frameshift_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_effect_on_frame
EVT-P-0007108-T01-IM5-79	P-0007108-T01-IM5		Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	CD74	in-frame	three_prime	117646132	34	1853	True	1853	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1 to 6 fused with ROS1 exons 34 to 43) : t(5;6)(chr5q32;chr6q22.1)(chr5:g.149783704::chr6:g.117646132) Protein fusion: in frame (CD74-ROS1)	NA
EVT-P-0027196-T01-IM6-80	P-0027196-T01-IM6		Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	CD74	out-of-frame	three_prime	117650377	33	1789	True	1789	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-7 fused to ROS1 exons 33-43) : t(5;6)(q32;q22.1)(chr5:g.149782321::chr6:g.117650377) Protein Fusion: out of frame (CD74-ROS1)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0026589-T02-IM6-81	P-0026589-T02-IM6		Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	CD74	in-frame	three_prime	117646032	34	1853	True	1853	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) Fusion (CD74 exon6 fused to ROS1 exon34) : t(5;6)(q32;q22.1)(chr5:g.149782990::chr6:g.117646032) Protein Fusion: in frame {CD74:ROS1}	NA
EVT-P-0021054-T01-IM6-82	P-0021054-T01-IM6		Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	CD74	in-frame	three_prime	117647162	34	1853	True	1853	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-6 fused to ROS1 exons 34-43): t(5;6)(q32;q22.1)(chr5:g.149783021::chr6:g.117647162) Protein Fusion: in frame {CD74:ROS1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0008134-T01-IM5-83	P-0008134-T01-IM5		Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	CD74	in-frame	three_prime	117646080	34	1853	True	1853	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		CD74 (NM_001025159) - ROS1 (NM_002944) fusion (CD74 exons 1-6 fused to ROS1 exons 34-43): t(5;6)(5q32;q22.1)(chr5:g.149783121::chr6:g.117646080) Protein fusion: in frame (CD74-ROS1)	NA
EVT-P-0023617-T01-IM6-84	P-0023617-T01-IM6		Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	CD74	in-frame	three_prime	117646444	34	1853	True	1853	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		CD74 (NM_00102515) - ROS1 (NM_002944) fusion (CD74 exons 1-6 fused to ROS1 exons 34-43): t(5;6)(q32;q22.1)(chr5:g.149782946::chr6:g.117646444) Protein Fusion: in frame {CD74:ROS1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frameshift	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0065104-T01-IM7-86	P-0065104-T01-IM7		Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	CD74	in-frame	five_prime	117646386	33	1853	True	1852	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False	Fibronectin type III domain	Protein kinase domain		ROS1 (NM_002944) - CD74 (NM_001025159) rearrangement: t(5;6)(q32;q22.1)(chr5:g.149783694::chr6:g.117646386) Protein Fusion: in frame {ROS1:CD74}	NA
EVT-P-0041118-T07-IM7-87	P-0041118-T07-IM7		Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	CD74	out-of-frame	five_prime	117648241	32	1789	True	1788	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False	Fibronectin type III domain	Protein kinase domain		ROS1 (NM_002944) - CD74 (NM_001025159) rearrangement: t(5;6)(q32;q22.1)(chr5:g.149782213::chr6:g.117648241) Protein Fusion: out of frame {ROS1:CD74}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frameshift	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_effect_on_frame
EVT-P-0008996-T04-IM6-88	P-0008996-T04-IM6		Salivary Gland Cancer	OSACA	ROS1-CEP85L	CEP85L	unknown	three_prime	117687428	18	875	False	875	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		CEP85L (NM_001178035) - ROS1 (NM_002944) rearrangement : c.242-19954:CEP85L_c.2623:ROS1del Protein Fusion: mid-exon {CEP85L:ROS1}	NA
EVT-P-0055135-T03-IM7-90	P-0055135-T03-IM7		Skin Cancer, Non-Melanoma	EMPD	ROS1-COL13A1	COL13A1	in-frame	five_prime	117658665	30	1693	True	1693	True	exact match	lost	0.0	False	Fibronectin type III domain	Protein kinase domain		ROS1 (NM_002944) - COL13A1 (NM_001130103) rearrangement: t(6;10)(q22.1;q22.1)(chr6:g.117658665::chr10:g.71715849) Protein Fusion: in frame {ROS1:COL13A1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0020458-T01-IM6-98	P-0020458-T01-IM6		Non-Small Cell Lung Cancer	LUAD	ROS1-EZR	EZR	in-frame	three_prime	117646033	34	1853	True	1853	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		EZR (NM_003379) - ROS1 (NM_002944) fusion (EZR exons 1-9 fused with ROS1 exons 34-43): c.1090+207:EZR_c.5558-455:ROS1del Protein Fusion: in frame {EZR:ROS1}	NA
EVT-P-006047-T02-IM5-99	P-006047-T02-IM5		Non-Small Cell Lung Cancer	LUAD	ROS1-EZR	EZR	in-frame	three_prime	117647241	34	1853	True	1853	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		EZR (NM_003379) - ROS1 (NM_002944) fusion (EZR exons 1-9 with ROS1 exons 34-43): c.1091-355:EZR_c.5557+146:ROS1del Protein fusion: in frame (EZR-ROS1)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000117-T01-IM3-100	P-000117-T01-IM3		Non-Small Cell Lung Cancer	LUAD	ROS1-EZR	EZR	unknown	three_prime	117650595	32	1755	False	1755	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		EZR (NM_003379) - ROS1 (NM_002944) Deletion: c.1090+214_c.5263del Protein fusion: mid-exon (EZR-ROS1)	NA
EVT-P-0010989-T01-IM5-101	P-0010989-T01-IM5		Non-Small Cell Lung Cancer	LUAD	ROS1-EZR	EZR	out-of-frame	three_prime	117649729	33	1789	True	1789	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		EZR (NM_003379) - ROS1 (NM_002944) fusion (EZR exons 1-9 fused with ROS1 exons 33-43): c.1090+215:EZR_c.5366+763:ROS1del Protein fusion: out of frame (EZR-ROS1)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0006047-T03-IM6-102	P-0006047-T03-IM6		Non-Small Cell Lung Cancer	NSCLCPD	ROS1-EZR	EZR	in-frame	three_prime	117647241	34	1853	True	1853	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		EZR (NM_003379) - ROS1 (NM_002944) fusion: c.1091-356:EZR_c.5557+146:ROS1del Protein Fusion: in frame {EZR:ROS1}	NA
EVT-P-0003970-T01-IM5-103	P-0003970-T01-IM5		Non-Small Cell Lung Cancer	LUAD	ROS1-EZR	EZR	out-of-frame	three_prime	117649698	33	1789	True	1789	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		ROS1 (NM_002944) - EZR (NM_003379) rearrangement: c.1090+142C: EZR_c.5366+794T:ROS1del Protein fusion: out of frame (EZR-ROS1)	NA
EVT-P-0014437-T01-IM6-104	P-0014437-T01-IM6		Non-Small Cell Lung Cancer	LUAD	ROS1-EZR	EZR	out-of-frame	three_prime	117650363	33	1789	True	1789	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		ROS1 (NM_002944) _EZR (NM_003379) rearrangement: c.1090+229:EZR_c.5366+129:ROS1del Protein fusion: out of frame (EZR-ROS1)	NA

event_id	sample_id	patient_id	tumor_type	oncotre_e_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0034609-T01-IM6-105	P-0034609-T01-IM6		Non-Small Cell Lung Cancer	LUAD	ROS1-EZR	EZR	in-frame	three_prime	117647244	34	1853	True	1853	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		EZR (NM_003379) - ROS1 (NM_002944) fusion: c. 1090+175:EZR_c.5557+143:ROS1del Protein Fusion: in frame {EZR:ROS1}	NA
EVT-P-0014304-T02-IM6-106	P-0014304-T02-IM6		Non-Small Cell Lung Cancer	LUAD	ROS1-EZR	EZR	in-frame	three_prime	117645949	34	1853	True	1853	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		EZR (NM_003379) - ROS1 (NM_002944) fusion: (EZR exons 1 to 9 fused with ROS1 exons 34 to 43); c. 1090+79:EZR_c.5558-371:ROS1del Protein Fusion: in frame {EZR:ROS1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_effect_on_frame
EVT-P-000445-T01-IM5-107	P-000445-T01-IM5		Ovarian Cancer	SBOV	ROS1-GOLGB1	GOLGB1	in-frame	three_prime	117645253	35	1881	True	1881	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		GOLGB1 (NM_004487) - ROS1 (NM_002944) rearrangement : t(3;6)(3q13.33;6q22.1)(chr3:121434446::chr6:117645253) Protein fusion: in frame (GOLGB1-ROS1)	NA
EVT-P-0005086-T01-IM5-108	P-0005086-T01-IM5		Glioma	GB	ROS1-GOPC	GOPC	in-frame	three_prime	117644378	35	1881	True	1881	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		GOPC(NM_020399) - ROS1(NM_002944) - Fusion (GOPC exon 8 fused to ROS1 exon 34) : c.1259-1533: GOPC_c.5641+1117:ROS1del Protein fusion: in frame (GOPC-ROS1)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0007626-T01-IM5-109	P-0007626-T01-IM5		Cancer of Unknown Primary	CUP	ROS1-GOPC	GOPC	in-frame	three_prime	117644860	35	1881	True	1881	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		GOPC (NM_020399) - ROS1 (NM_002944) fusion (GOPC exons 1-8 fused with ROS1 exons 35-43) : c.1259-1217:GOPC_c.5641+635:ROS1del Protein fusion: in frame (GOPC-ROS1)	NA
EVT-P-0019111-T01-IM6-111	P-0019111-T01-IM6		Non-Small Cell Lung Cancer	LUAD	ROS1-GOPC	GOPC	out-of-frame	five_prime	117643452	34	1881	True	1880	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False	Fibronectin type III domain	Protein kinase domain		ROS1 (NM_002944) - GOPC (NM_020399) rearrangement: c.5642-895:ROS1_c.912+138:GOPCdup Protein Fusion: out of frame {ROS1:GOPC}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frameshift	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0058302-T02-IM7-112	P-0058302-T02-IM7		Breast Cancer	IDC	ROS1-GRK2	GRK2	unknown	five_prime	117622188	42	2228	False	2227	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	Protein kinase domain; Fibronectin type III domain			ROS1 (NM_002944) - GRK2 (NM_021956) fusion: c. 6682:ROS1_c.2086-17008:GRK2inv Protein Fusion: mid-exon {ROS1:GRK2}	NA
EVT-P-0021914-T01-IM6-113	P-0021914-T01-IM6		Soft Tissue Sarcoma	MFH	ROS1-LARS1	LARS1	out-of-frame	five_prime	117646519	33	1853	True	1852	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False	Fibronectin type III domain	Protein kinase domain		ROS1 (NM_002944) - LARS (NM_020117) rearrangement: t(5;6)(q32;q22.1)(chr5:g.145500267::chr6:g.117646519) Protein Fusion: out of frame {ROS1:LARS}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0036702-T01-IM6-118	P-0036702-T01-IM6		Breast Cancer	IDC	ROS1-NUS1	NUS1	unknown	three_prime	117711069	12	421	True	421	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		ROS1 (NM_002944) rearrangement: c.1263-60:ROS1_chr6:g.118032766inv Transcript Fusion {NUS1:ROS1}	NA
EVT-P-016046-T01-IM6-171	P-0016046-T01-IM6		Non-Small Cell Lung Cancer	LUAD	ROS1-SLC34A2	SLC34A2	out-of-frame	three_prime	117662059	31	1694	True	1694	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		ROS1 (NM_002944) - SLC34A2 (NM_006424) fusion: t(4;6)(p15;q22)(chr4:g.25667671::chr6:g.117662059) Protein Fusion: out of frame {SLC34A2:ROS1}	NA
EVT-P-010984-T01-IM5-175	P-0010984-T01-IM5		Pancreatic Cancer	PAAD	ROS1-SLC4A4	SLC4A4	in-frame	three_prime	117646719	34	1853	True	1853	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		SLC4A4 (NM_001134742) - ROS1 (NM_002944) rearrangement: t(4;6)(q13.3;q22.1)(chr4:g.72429843::chr6:g.117646719) Protein fusion: in frame (SLC4A4-ROS1)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0013451-T01-IM5-178	P-0013451-T01-IM5		Melanoma	SKCM	ROS1-STX7	STX7	out-of-frame	three_prime	117649456	33	1789	True	1789	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		STX7 (NM_003569) - ROS1 (NM_002944) rearrangement: c.86-3411:STX7_c.5366+1036:ROS1del Protein fusion: out of frame (STX7-ROS1)	NA
EVT-P-0019670-T03-IM6-180	P-0019670-T03-IM6		Soft Tissue Sarcoma	IMT	ROS1-TFG	TFG	in-frame	three_prime	117643761	35	1881	True	1881	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		TFG (NM_001195478) - ROS1 (NM_002944) fusion (TFG exon 4 fused to ROS1 exon 35): t(3;6)(q12.2;q22.1)(chr3:g.100448554::chr6:g.117643761) Protein Fusion: in frame {TFG:ROS1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0019670-T02-IM6-181	P-0019670-T02-IM6		Soft Tissue Sarcoma	IMT	ROS1-TFG	TFG	in-frame	three_prime	117643761	35	1881	True	1881	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		TFG (NM_001195478) - ROS1 (NM_002944) fusion (TFG exon 4 fused to ROS1 exon 35): t(3;6)(q12.2;q22.1)(chr3:g.100448554::chr6:g.117643761) Protein Fusion: in frame {TFG:ROS1}	NA
EVT-P-0003207-T01-IM5-183	P-0003207-T01-IM5		Non-Small Cell Lung Cancer	LUAD	ROS1-TPM3	TPM3	in-frame	three_prime	117642993	35	1881	True	1881	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		NA Protein fusion: in frame (TPM3-ROS1)	NA
EVT-P-0028865-T02-IM6-184	P-0028865-T02-IM6		Non-Small Cell Lung Cancer	LUAD	ROS1-TPM3	TPM3	in-frame	three_prime	117643589	35	1881	True	1881	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		TPM3 (NM_001278190) - ROS1 (NM_002944) fusion (TPM3 exons 1-6 with ROS1 exons 35-43): t(1;6)(q21.3;q22.1)(chr1:g.154133648::chr6:g.117643589) Protein Fusion: in frame {TPM3:ROS1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0001840-T01-IM3-185	P-0001840-T01-IM3		Non-Small Cell Lung Cancer	LUAD	SDC4-ROS1	SDC4	in-frame	three_prime	117650688	32	1750	True	1750	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		SDC4 (NM_002999) - ROS1 (NM_002944) Translocation: t(20;6)(q13.12;q22.1)(chr20:43962892::chr6:117650688) Protein fusion: in frame (SDC4-ROS1)	NA
EVT-P-0001840-T02-IM5-186	P-0001840-T02-IM5		Non-Small Cell Lung Cancer	LUAD	SDC4-ROS1	SDC4	in-frame	three_prime	117650732	32	1750	True	1750	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		SDC4 (NM_002999) - ROS1 (NM_002944) fusion: t(20;6)(q13.12;q22.1)(chr20:g.43962893::chr6:g.117650732) Protein fusion: in frame (SDC4-ROS1)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0054419-T02-IM6-187	P-0054419-T02-IM6		Non-Small Cell Lung Cancer	LUAD	SDC4-ROS1	SDC4	unknown	three_prime	117645734	34	1853	True	1853	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		SDC4 (NM_002999) - ROS1 (NM_002944) fusion: t(6;20)(q22.1;q13.12)(chr6:g.117645734::chr20:g.43955928) Protein Fusion: mid-exon {SDC4:ROS1}	NA
EVT-P-0054419-T01-IM6-188	P-0054419-T01-IM6		Non-Small Cell Lung Cancer	LUAD	SDC4-ROS1	SDC4	unknown	three_prime	117645734	34	1853	True	1853	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		SDC4 (NM_002999) - ROS1 (NM_002944) fusion: t(6;20)(q22.1;q13.12)(chr6:g.117645734::chr20:g.43955928) Protein Fusion: mid-exon {SDC4:ROS1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frameshift	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0047840-T02-IM6-189	P-0047840-T02-IM6		Non-Small Cell Lung Cancer	LUAD	SDC4-ROS1	SDC4	unknown	three_prime	117645871	34	1853	True	1853	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		SDC4 (NM_002999) - ROS1 (NM_002944) fusion: t(6;20)(q22.1;q13.12)(chr6:g.117645871::chr20:g.43955947) Protein Fusion: mid-exon {SDC4:ROS1}	NA
EVT-P-004414-T01-IM5-190	P-004414-T01-IM5		Non-Small Cell Lung Cancer	LUAD	SDC4-ROS1	SDC4	in-frame	three_prime	117651952	32	1750	True	1750	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		SDC4 (NM_002999) - ROS1 (NM_002944) fusion (SDC4 exons 1-4 fused with ROS1 exons 32-43): t(6;20)(q22.1;q13.12)(chr6:g.117651952::chr20:g.43957039) Protein fusion: in frame (SDC4-ROS1)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_effect_on_frame
EVT-P-0043938-T01-IM6-191	P-0043938-T01-IM6		Non-Small Cell Lung Cancer	LUAD	SDC4-ROS1	SDC4	in-frame	three_prime	117651208	32	1750	True	1750	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		SDC4 (NM_002999) - ROS1 (NM_002944) fusion: t(6;20)(q22.1;q13.12)(chr6:g.117651208::chr20:g.43962372) Protein Fusion: in frame {SDC4:ROS1}	NA
EVT-P-0004946-T02-IM5-192	P-0004946-T02-IM5		Non-Small Cell Lung Cancer	LUAD	SDC4-ROS1	SDC4	in-frame	three_prime	117651013	32	1750	True	1750	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		SDC4 (NM_002999) - ROS1 (NM_002944) fusion: t(6;20)(q22.1;q13.12)(chr6:g.117651013::chr20:g.43962011) Protein fusion: in frame (SDC4-ROS1)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0038394-T01-IM6-193	P-0038394-T01-IM6		Non-Small Cell Lung Cancer	LUAD	SDC4-ROS1	SDC4	in-frame	three_prime	117654360	32	1750	True	1750	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		SDC4 (NM_002999) - ROS1 (NM_002944) rearrangement: t(6;20)(q22.1;q13.12)(chr6:g.117654360::chr20:g.43961742) Protein Fusion: in frame {SDC4:ROS1}	NA
EVT-P-0027630-T01-IM6-194	P-0027630-T01-IM6		Non-Small Cell Lung Cancer	LUAD	SDC4-ROS1	SDC4	in-frame	three_prime	117651891	32	1750	True	1750	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		SDC4 (NM_002999) - ROS1 (NM_002944) fusion (SDC4 exons 1-2 fused to ROS1 exons 32-43): t(6;20)(q22.1;q13.12)(chr6:g.117651891::chr20:g.43962429) Protein Fusion: in frame {SDC4:ROS1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0064541-T01-IM7-195	P-0064541-T01-IM7		Non-Small Cell Lung Cancer	LUAD	SDC4-ROS1	SDC4	in-frame	three_prime	117650669	32	1750	True	1750	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		SDC4 (NM_002999) - ROS1 (NM_002944) fusion: t(6;20)(q22.1;q13.12)(chr6:g.117650669::chr20:g.43962537) Protein Fusion: in frame {SDC4:ROS1}	NA
EVT-P-0022578-T02-IM6-196	P-0022578-T02-IM6		Non-Small Cell Lung Cancer	LUAD	SDC4-ROS1	SDC4	in-frame	three_prime	117652158	32	1750	True	1750	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		SDC4 (NM_002999) - ROS1 (NM_002944) fusion (SDC4 exons 1-2 fused in-frame to ROS1 exons 32-43): t(6;20)(q22.1;q13.12)(chr6:g.117652158::chr20:g.43962266) Protein Fusion: in frame {SDC4:ROS1}	NA

event_id	sample_id	patient_id	tumor_type	oncotre_e_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0022578-T01-IM6-197	P-0022578-T01-IM6		Non-Small Cell Lung Cancer	LUAD	SDC4-ROS1	SDC4	in-frame	three_prime	117652158	32	1750	True	1750	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		SDC4 (NM_002999) - ROS1 (NM_002944) fusion (SDC4 exons 1-2 fused in-frame to ROS1 exons 32-43): t(6;20)(q22.1;q13.12)(chr6:g.117652158::chr20:g.43962266) Protein Fusion: in frame {SDC4:ROS1}	NA
EVT-P-0012008-T01-IM5-198	P-0012008-T01-IM5		Non-Small Cell Lung Cancer	LUAD	SDC4-ROS1	SDC4	in-frame	three_prime	117652119	32	1750	True	1750	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		SDC4 (NM_002999) - ROS1 (NM_002944) reciprocal fusion (SDC4 exons 1-2 with ROS1 exons 32-43): t(6;20)(q22.1;q13.12) (chr6:g.117652119::chr20:g.43964016) Protein fusion: in frame (SDC4-ROS1)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-017325-T01-IM6-199	P-0017325-T01-IM6		Non-Small Cell Lung Cancer	LUAD	SDC4-ROS1	SDC4	in-frame	three_prime	117654375	32	1750	True	1750	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		SDC4 (NM_002999) - ROS1 (NM_002944) fusion (SDC4 exons 1-2 fused in-frame to ROS1 exons 32-43): t(6;20)(q22.1;q13.12)(chr6:g.117654375::chr20:g.43963860) Protein Fusion: in frame {SDC4:ROS1}	NA
EVT-P-012008-T02-IM6-200	P-0012008-T02-IM6		Non-Small Cell Lung Cancer	LUAD	SDC4-ROS1	SDC4	in-frame	three_prime	117652099	32	1750	True	1750	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		SDC4 (NM_002999) - ROS1 (NM_002944) reciprocal fusion (SDC4 exons 1-2 with ROS1 exons 32-43): t(6;20)(q22.1;q13.12) (chr6:g.117652119::chr20:g.43964016) Protein Fusion: in frame {SDC4:ROS1}	NA

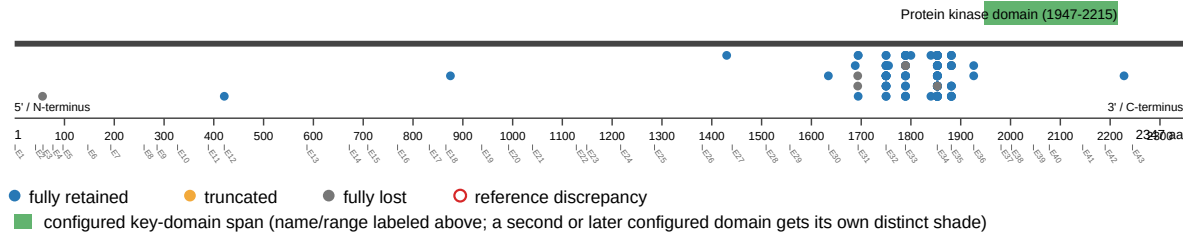
event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0016504-T01-IM6-201	P-0016504-T01-IM6		Non-Small Cell Lung Cancer	LUAD	SDC4-ROS1	SDC4	in-frame	three_prime	117650681	32	1750	True	1750	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		SDC4 (NM_002999) - ROS1 (NM_002944) fusion (SDC4 exons 1-2 fused in-frame to ROS1 exons 32-43): t(6;20)(q22.1;q13.12)(chr6:g.117650681::chr20:g.43961904) Protein Fusion: in frame {SDC4:ROS1}	NA
EVT-P-0003507-T01-IM5-202	P-0003507-T01-IM5		Bladder Cancer	BLCA	SLC16A10-ROS1	SLC16A10	in-frame	five_prime	117739561	2	56	True	56	True	exact match	lost	0.0	False		Protein kinase domain; Fibronectin type III domain		NA Protein fusion: in frame (ROS1-SLC16A10)	NA
EVT-P-0042481-T01-IM6-203	P-0042481-T01-IM6		Cancer of Unknown Primary	ADNOS	SLC4A4-ROS1	SLC4A4	out-of-frame	three_prime	117646662	34	1853	True	1853	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		SLC4A4 (NM_001134742) - ROS1 (NM_002944) fusion: t(4;6)(q13.3;q22.1)(chr4:g.72430718::chr6:g.117646662) Protein Fusion: out of frame {SLC4A4:ROS1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_effect_on_frame
EVT-P-0045322-T02-IM6-205	P-0045322-T02-IM6		Soft Tissue Sarcoma	IMT	TFG-ROS1	TFG	in-frame	three_prime	117644683	35	1881	True	1881	True	exact match	retained	1.0	False	Protein kinase domain	Fibronectin type III domain		TFG (NM_001195478) - ROS1 (NM_002944) fusion: t(3;6)(q12.2;q22.1)(chr3:g.100448076::chr6:g.117644683) Protein Fusion: in frame {TFG:ROS1}	NA
EVT-P-0007870-T02-IM6-207	P-0007870-T02-IM6		Non-Small Cell Lung Cancer	LUAD	ZNF157-ROS1	ZNF157	unknown	five_prime	117659405	30	1693	True	1693	True	exact match	lost	0.0	False	Fibronectin type III domain	Protein kinase domain		ROS1 (NM_002944) Rearrangement : t(X;6)(p11.23;q22.1)(chrX:g.47262831; chr6:g.117659405) Antisense Fusion	NA

Figures

domain retention outliers

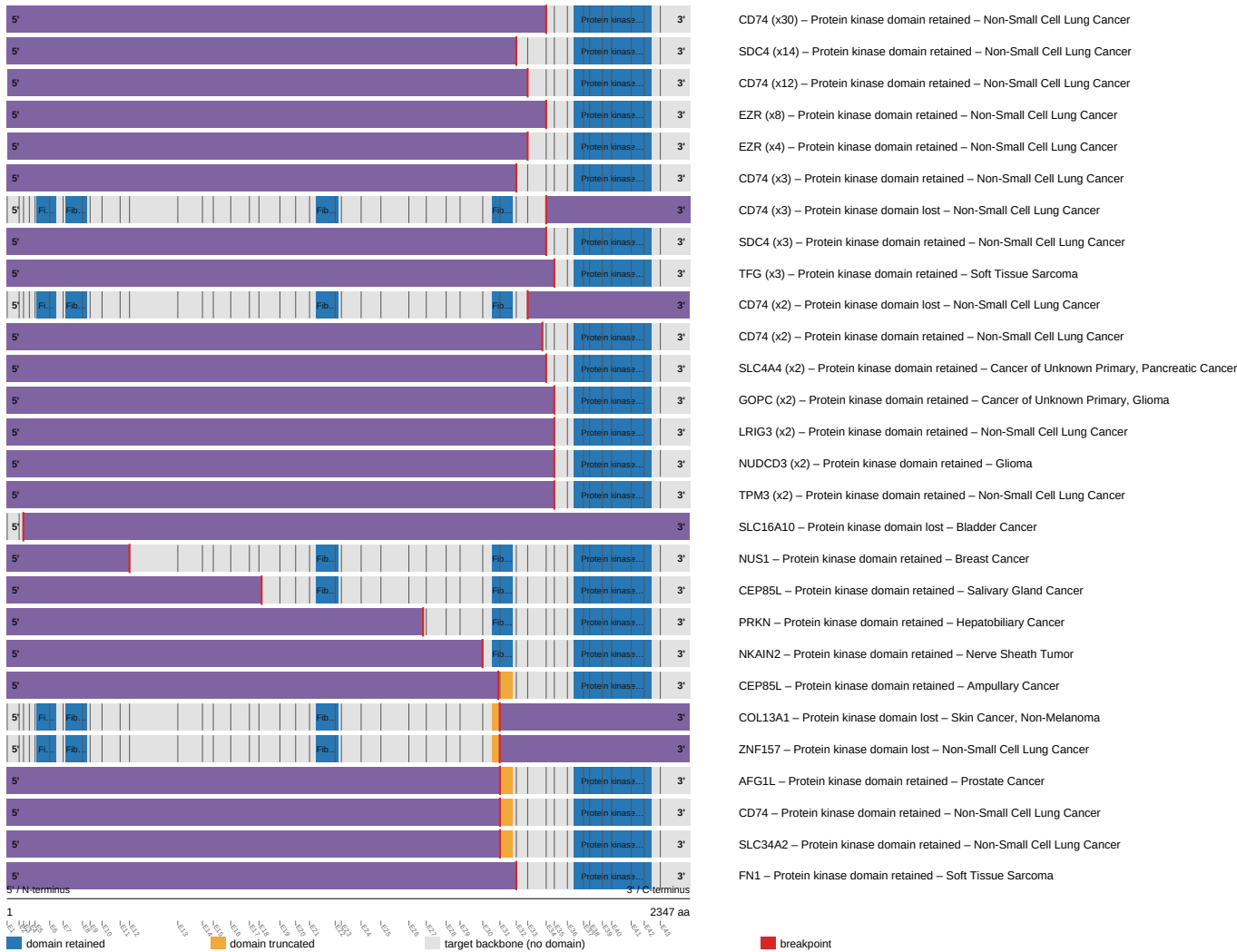
ROS1 domain-retention track



fusion schematic

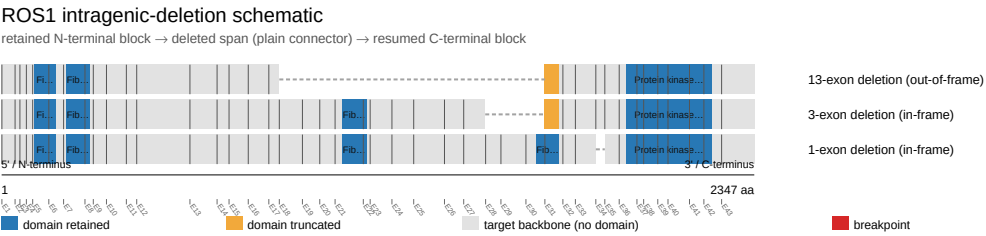
ROS1 fusion-transcript schematic

Fusion order varies by row; follow each row's 5' and 3' labels (ROS1 portion is domain-colored, exon ticks). Partner-gene blocks share one color; row labels identify each partner.



Partner-block colors are decorative; row labels identify partners.
Showing the top 28 of 40 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

intragenic deletion schematic



reference comparison

Reference vs msk_impact_50k_2026

